LLM-Guided ANN Index Optimization for Efficient Human-Object Interaction Retrieval

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ABSTRACT

Modern vector database management systems serve large-scale semantic search by combining ANN index retrieval with multi-stage reranking pipelines — exposing a configuration space of graph connectivity, search depth, candidate pool size, and cross-modal fusion weight that jointly governs the accuracy–throughput trade-off. Determining the right values for these parameters is a long-standing challenge: grid and random search evaluate every candidate with no memory of prior outcomes, while Bayesian methods model each parameter independently and fail to capture the cross-stage coupling where Stage-1 search depth and Stage-2 fusion weight must be tuned jointly — a structural dependency that existing techniques do not model.

Hence, we propose a phase-aware LLM agent that conditions each configuration proposal on the complete history of measured outcomes, reasoning over cross-stage parameter dependencies. Firstly, we design a three-phase optimization schedule — exploration, exploitation, and fine-tuning — with stagnation detection to prevent premature convergence. Secondly, to evaluate on a semantically rich and realistic retrieval workload, we (1) construct the first text-query image retrieval benchmark derived from HICO-DET on Intel's VDMS vector DBMS, (2) formally *prove* that CLIP text embeddings occupy the semantic centroid of each HOI category's visual cluster — making text queries *structurally* superior to image queries on compositionally structured data — and (3) build a zero-cost DINOv2 pseudo-relevance-feedback reranking stage that substantially improves retrieval quality at negligible latency overhead. Finally, we evaluate across three datasets of increasing parameter-space simplicity to characterize when LLM-guided reasoning provides the greatest advantage over classical methods.

Compared to classical and learned optimizers, the agent achieves +9.7% accuracy–throughput Score over Optuna TPE and +3.3% over GP-BO (a VDTuner-style Gaussian Process surrogate) on HICO-DET, converging in ∼13 iterations on average versus ∼20 for Optuna TPE. On a large-scale cross-domain benchmark, the agent reaches near-optimal quality in ∼1–2 iterations versus ∼13 for Optuna TPE, with performance gaps narrowing on simpler spaces — validating that the agent's advantage scales with the coupling complexity of the configuration landscape.

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1 INTRODUCTION

Modern vector database management systems expose a multidimensional configuration space whose parameters jointly govern the accuracy–throughput trade-off at production scale. A system such as VDMS [31], Milvus [32], or GaussDB-Vector [29] serving semantic similarity queries over tens of millions of embeddings requires setting graph connectivity (M), search depth (*efSearch*), candidate pool size (k), cross-modal fusion weight α, reranker reference count (n_{refs}), and reference selection strategy — a 7-parameter space in which each evaluation requires a full index rebuild and query execution across tens of thousands of embeddings, incurring significant wall-clock cost per trial [36] — making convergence speed a first-class system concern. The interactions among parameters are highly non-linear: the optimal value of each parameter depends on the values of the others in ways that resist independent analysis.

State-of-the-art approaches automate the search via grid enumeration, random sampling, or Bayesian optimization [3, 27]. These methods follow a trial-and-error strategy with no cross-parameter memory: grid and random search evaluate every candidate independently, while Bayesian methods such as Optuna TPE [1] model each parameter with independent marginal distributions [3]. This independence assumption is harmless on scalar parameter spaces, but it causes *structural* failure on multi-stage retrieval pipelines, where Stage-1 search depth (*efSearch*) and Stage-2 fusion weight (α) are jointly determined: reducing *efSearch* compresses the candidate pool fed to the reranker, an effect that can only be compensated by raising α. No parameter-independent technique models this inter-stage coupling, causing all three method families to systematically miss compensatory operating points — configurations

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that are globally optimal yet structurally invisible to blind optimizers regardless of iteration budget.

The core challenge is therefore: *how do we build an optimizer that reasons over the full history of (configuration, Score) pairs while respecting conditional parameter dependencies and cross-stage coupling?* A necessary precondition is a benchmark with sufficient coupling complexity to distinguish optimizer sophistication from random luck. Pure vector search datasets such as SIFT1M [10] and ANN-Benchmarks [2] expose a small, near-independent parameter space where any competent optimizer converges within tens of iterations. We address this gap by constructing the first text-query image retrieval benchmark derived from HICO-DET [4] — 47,776 images across 600 fine-grained verb–object interaction categories, deployed on Intel's VDMS vector DBMS — whose compositional structure generates a retrieval workload rich enough to expose the structural limits of blind ANN optimizers. Every prior use of HICO-DET has been restricted to the detection paradigm [12, 16, 23, 38]; we identify and fill the orthogonal retrieval direction. To evaluate on this benchmark we further construct a two-stage retrieval pipeline pairing CLIP ViT-L/14 ANN retrieval with zero-cost DINOv2 ViT-L/14-reg4 pseudo-relevance-feedback reranking, and formally prove that CLIP text embeddings occupy the semantic centroid of each HOI category's visual cluster — a geometric result that makes text queries structurally superior to image queries on compositionally structured data and directly justifies the reranker design without a secondary index.

Two parallel threads of prior work address orthogonal aspects of this problem, yet neither reaches the intersection our setting requires. On the *LLM-guided tuning* side, GPTuner [13], λ-Tune [9] (SIGMOD 2025), and OtterTune [30] show that LLM reasoning and learning-based methods outperform Bayesian and reinforcement-learning baselines on DBMS configuration — but all three operate on flat, scalar knob spaces (PostgreSQL, MySQL) where parameters are conditionally independent and no cross-stage coupling exists by construction. On the *ANN parameter tuning* side, VDTuner [36] applies multi-objective Bayesian optimization to VDMS index parameters, and VSAG [40] (VLDB 2025) eliminates index rebuilds via analytical subgraph monotonicity — but both treat parameters independently and neither models the coupling between *efSearch* and α that emerges when ANN retrieval feeds a reranker. Our work occupies the intersection that neither thread reaches: the first LLM-guided optimizer designed specifically for ANN index spaces with hard conditional type-dependencies and cross-stage coupling, reasoning over the joint efSearch–α trade-off that all parameter-independent methods structurally cannot model, regardless of iteration budget or LLM capability.

To address these challenges, we propose a *phase-aware LLM agent* that conditions each configuration proposal on the complete history of measured (configuration, Score) pairs, reasoning explicitly over cross-stage parameter dependencies across three self-regulated optimization phases — exploration, exploitation, and fine-tuning — with stagnation detection and anti-collapse mechanisms. The agent is deployed on Intel's VDMS and evaluated across three datasets of increasing parameter-space simplicity: HICO-DET (7-parameter, cross-modal, full cross-stage coupling), Google Landmarks v2 [34] (GLDv2, 762K images, image-to-image retrieval), and SIFT1M [10] (4-parameter, pure ANN search). Compared to state-of-the-art baselines, the agent achieves +9.7% accuracy–throughput Score over Optuna TPE and +3.3% over GP-BO (a VDTuner-style Gaussian Process surrogate) on HICO-DET, converging in ~13 iterations on average versus ~20 for Optuna TPE — a 2.6× improvement over the CLIP FaissFlat baseline. On GLDv2, the agent reaches near-optimal quality in ~1–2 iterations versus ~13 for Optuna TPE. On SIFT1M, all methods converge within 2% of each other — confirming that the agent's advantage scales with the structural coupling of the configuration landscape.

Contributions. This paper makes the following contributions:

(1) A phase-aware LLM agent for ANN hyperparameter optimization. We identify that ANN retrieval search spaces have hard conditional dependencies — engine type gates valid parameters, and Stage-1 search depth couples to Stage-2 fusion weight — that cause parameter-independent methods to fail structurally on this class of problems. We present a phase-aware LLM agent that conditions each proposal on the full history of (configuration, Score) pairs, discovering the compensatory operating point efSearch=64 $<$ k=200 with α=0.80 that neither Bayesian nor random methods locate within 30 iterations. The agent converges to ~95% of its best Score in ~13 iterations on average versus ~20 for Optuna TPE, and achieves a mean Score = 4.44 (mAP×QPS) across seeds — a 2.6× improvement over the CLIP FaissFlat baseline and 13.9× over brute-force CLIP+DINOv2 reranking. Mechanistic ablations confirm that *both* components are independently necessary: removing history conditioning degrades mean Score by 3.2% and removing phase-aware transitions by 5.2%, with neither ablation locating the compensatory efSearch=64, α=0.80 operating point across any seed.

(2) The first HOI retrieval benchmark. We construct a text-query image retrieval benchmark from HICO-DET (47,776 images, 600 compositional HOI categories, 90,641 ground-truth pairs), deployed on Intel's VDMS, providing a coupled configuration landscape rich enough to distinguish optimizer sophistication from random luck. The benchmark's fine-grained verb–object structure creates a retrieval task qualitatively harder than scene- or object-level benchmarks: even exhaustive exact-search with modality fusion reaches only mAP = 0.276, exposing a semantic ceiling that reflects language–vision alignment difficulty rather than any indexing limitation.

(3) A deployment-efficient two-stage retrieval pipeline. We propose pairing CLIP ViT-L/14 ANN retrieval with DINOv2 ViT-L/14-reg4 pseudo-relevance-feedback reranking at zero additional indexing cost, achieving +3.65 pp mAP (+15.2% relative) over ANN-only retrieval at under 1% additional query latency, with a single ANN index at deployment.

(4) A geometric theory of text query superiority. We formally prove that a CLIP text embedding occupies the semantic centroid of its HOI category's visual cluster, while any reference image is displaced from that centroid by $\varepsilon_j > 0$, contributing a bias term ε_j^2 to the expected distance from every class member. This centroid-displacement theory justifies the zero-cost reranker design — averaging n_r centroid-proximal reference images minimizes ε_j without a secondary index — and predicts the systematic mAP advantage of text queries over image queries on compositionally structured datasets.

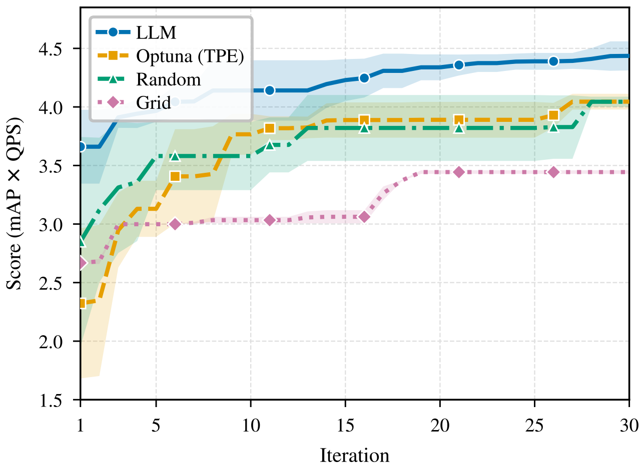

Figure 1: Motivating example: LLM-guided optimization converges faster than all baselines on HICO-DET. Best Score (mAP × QPS) achieved so far versus iteration (mean $\pm 1\sigma$, 3 seeds). The LLM agent reaches Score ≈ 3.97 at iteration 6 — a level that Optuna TPE first matches at iteration 27 and that GP-BO and Random Search do not exceed within 30 iterations at comparable scores. The advantage stems from explicit reasoning over cross-stage parameter dependencies (k–α coupling) that parameter-independent methods structurally cannot model. Full empirical breakdown in Section 6.

(5) A complexity-matched evaluation across three retrieval workloads. We evaluate across HICO-DET, GLDv2, and SIFT1M — datasets of systematically increasing parameter-space simplicity — to characterize when and why LLM-guided reasoning outperforms classical methods. On HICO-DET the agent achieves +9.7% over Optuna TPE and +3.3% over GP-BO (VDTuner-style); on GLDv2 it reaches near-optimal quality in ~1–2 iterations versus ~13 for Optuna TPE; on SIFT1M all methods converge within 2%. This graded evaluation establishes that the agent's advantage is proportional to the structural coupling of the configuration landscape, providing a principled framework for predicting optimizer behavior across retrieval workloads.

The remainder of this paper is organized as follows. Section 2 reviews related work on HOI understanding, image retrieval, vector database systems, and hyperparameter optimization. Section 3 describes the construction of the HICO-DET retrieval benchmark. Section 4 presents the two-stage retrieval pipeline and the embedding geometry analysis. Section 5 details the LLM-guided ANN hyperparameter optimizer. Section 6 reports experimental results and ablations. Section 7 concludes.

2 RELATED WORK

2.1 Human-Object Interaction Detection

HOI detection has been studied extensively since the introduction of HICO-DET [4], which defined the two-stage paradigm of detecting human–object pairs and classifying their interactions across 600 fine-grained verb–object categories. Early approaches modeled spatial and appearance relationships between detected instances via instance-centric attention [8]. The adoption of transformer architectures enabled end-to-end detection [12, 38], and subsequent work leveraged vision-language pre-training to transfer interaction semantics from large-scale models [16, 17, 23]. HOICLIP [23] (CVPR 2023) distills CLIP interaction semantics into a detector via knowledge transfer, while DP-HOI [16] (CVPR 2024) applies disentangled pre-training to reduce annotation dependence. Despite sustained progress on mean Average Precision for detection, every published use of HICO-DET is restricted to the detection paradigm — predicting bounding boxes and interaction labels from a given image. We identify and fill the orthogonal *retrieval* direction: given a natural-language HOI query, rank all 47,776 database images by relevance. The compositional verb–object label structure that makes HOI detection challenging creates an equally demanding retrieval workload, one with a 7-parameter coupled ANN configuration space that exposes the structural limits of blind optimizers.

2.2 Cross-Modal Image Retrieval

Text-image retrieval research established benchmark datasets from caption annotations [18, 37] and proposed contrastive embedding losses to bring matching pairs close in a shared space [5]. CLIP [25] transformed the field: its joint vision-language embedding enables zero-shot retrieval across diverse domains without task-specific fine-tuning. BLIP-2 [14] extended this paradigm by bridging frozen visual encoders and large language models via a lightweight querying transformer, achieving strong generalization with substantially fewer trainable parameters. UniIR [33] (ECCV 2024 Oral) unified eight multimodal retrieval tasks in a single instruction-tuned model, constructing the M-BEIR benchmark spanning 1.5M queries over 10 datasets; we adopt UniIR and standalone CLIP as primary baselines. LamRA [20] (CVPR 2025) repurposes generative large multimodal models for universal retrieval via lightweight LoRA adaptation, achieving strong cross-task generalization on M-BEIR without task-specific fine-tuning. None of these works constructs a retrieval benchmark from HOI annotations, analyzes the relative merits of text versus image queries on compositionally structured data, or addresses the downstream ANN index configuration problem that arises when deploying such a retrieval system in a production VDMS. We fill all three gaps.

2.3 Vector Database Management Systems and ANN Indexing

ANN index research has produced a rich family of algorithms: graph-based indices (HNSW [21], DiskANN [28]), inverted file structures with product quantization [10], and GPU-accelerated flat search [11]. HNSW [21] constructs a hierarchical proximity graph supporting $O(\log n)$-complexity search with tunable accuracy–throughput trade-offs governed by M and *efSearch*; DiskANN [28] extends billion-scale graph search to SSD-resident indices. ANN-Benchmarks [2] and the NeurIPS 2021 billion-scale competition [26] systematically characterize the Pareto frontier of recall versus throughput across these algorithms and datasets.

Production VDBMSs integrate ANN indexing with metadata management, persistent storage, and query planning. Milvus [32] supports multiple index backends with a unified query interface; VDMS [31] co-locates PMGD graph-property metadata with vector data, enabling structured pre-filters on ANN queries — a feature we exploit as the constraint_strategy optimization dimension. GaussDB-Vector [29] (VLDB 2025) addresses LLM-scale real-time ingestion by decoupling indexing from serving, supporting mixed query types at billion-vector scale. VSAG [40] (VLDB 2025) eliminates costly index rebuilds during parameter sweeps by exploiting analytical subgraph monotonicity, making parameter-search over graph indices significantly cheaper. Despite this infrastructure richness, *index configuration* — choosing engine type, graph connectivity M, search depth *efSearch*, candidate pool size k, and cross-stage fusion weight α for a given embedding distribution and throughput target — is universally left to blind grid or random search. Our LLM agent replaces these procedures with history-conditioned sequential reasoning over the full mixed-type parameter space.

2.4 Hyperparameter Optimization

HPO methods divide into three families. **Heuristic search:** Grid and random search [3] are the dominant baselines; random search is empirically superior in high-dimensional spaces because it covers more of the marginal space of each parameter. **Bayesian optimization (BO):** Gaussian-process-based BO [27] fits a probabilistic surrogate and acquires the next configuration by maximizing expected improvement; BOHB [6] hybridizes kernel density estimation with successive halving for additional efficiency. **Bandit and meta-learning:** Hyperband [15] treats configuration selection as a pure-exploration bandit problem; Auto-sklearn [7] applies BO to joint algorithm selection and hyperparameter configuration. ML-guided DBMS configuration was pioneered by OtterTune [30], which applied Gaussian process regression to map workload features to optimal knob settings; subsequent systems applied reinforcement learning, but all operate on continuous or integer knob spaces without structural type-dependencies.

Most closely related on the VDMS side, VDTuner [36] applies multi-objective Bayesian optimization to tune both index parameters and system parameters in Milvus, using a successive-abandon strategy to allocate the evaluation budget across index types with non-fixed parameter spaces, achieving up to 3.57× faster convergence than baselines. However, VDTuner evaluates on pure vector search benchmarks (GloVe, keyword-match) without semantic structure and models all parameters independently via a Gaussian process surrogate — it does not account for the cross-stage coupling between ANN search depth and reranker fusion weight that arises in multi-stage retrieval pipelines.

The fundamental limitation of all classical HPO methods for ANN index selection is that the search space is *mixed-type*: discrete engine choices (*HNSW* vs. *IVFFlat*), integer parameters (M, *efSearch*, *nlist*, k), and continuous parameters (α), with hard structural dependencies between parameters and engine choice. Gaussian process surrogates assume stationarity over a continuous space, making them ill-suited to this setting. Tree-structured Parzen Estimators (TPE), as implemented in Optuna [1], handle mixed-type spaces more gracefully by modeling the density of good and bad configurations separately, but they treat each parameter independently and structurally cannot exploit the knowledge that reducing *efSearch* compresses the candidate pool fed to a reranker — an effect that must be compensated by raising α. This independence assumption causes TPE to miss the compensatory operating point efSearch=64, α=0.80 even within a 30-iteration budget.

Table 1: Comparison of closely related configuration optimization methods. ✓/× indicate whether the property is present.

Method	LLM-Guided	Cross-Stage Coupling	Mixed-Type Space	Semantic Benchmark
OtterTune [30]	×	×	×	×
GPTuner [13]	✓	×	×	×
λ-Tune [9]	✓	×	×	×
VDTuner [36]	×	×	✓	×
Optuna TPE [1]	×	×	✓	×
Ours	✓	✓	✓	✓

2.5 LLM-Guided System Configuration

Recent work established that large language models can act as black-box optimizers by iteratively proposing candidate solutions and refining them based on observed outcomes. OPRO [35] encodes the history of (solution, score) pairs directly in the prompt, enabling the LLM to generate improved solutions without gradient information. Follow-up work applied this paradigm to machine learning hyperparameter optimization [19, 39], showing that LLMs match or surpass Bayesian optimization in sample efficiency on tabular HPO benchmarks by exploiting domain knowledge embedded in pre-training.

In the database configuration domain, GPTuner [13] (VLDB 2024) uses GPT-4 to extract structured priors from DBMS manuals and guide a coarse-to-fine Bayesian optimization of PostgreSQL and MySQL knobs, building on OtterTune [30] with substantially reduced tuning cost. λ-Tune [9] (SIGMOD 2025) harnesses LLMs for automated database system tuning, using LLM-generated workload analysis to warm-start configuration search and demonstrating that LLM reasoning transfers to heterogeneous knob spaces. Both systems operate on flat, scalar knob spaces (buffer pool sizes, parallelism settings, I/O parameters) where parameters are conditionally independent by construction: no engine type gates valid parameters, and no cross-stage coupling arises because DBMS knob tuning has a single-objective, single-stage evaluation loop.

Our work occupies the intersection that neither the HPO thread (VDTuner [36], Optuna [1], random search [3]) nor the LLM-guided DBMS tuning thread (GPTuner [13], λ-Tune [9], OtterTune [30]) reaches: a history-conditioned LLM agent designed specifically for ANN index spaces with *hard conditional type-dependencies* (engine type gates valid parameters) and *cross-stage coupling* (*efSearch* and α must be jointly optimized when ANN retrieval feeds a reranker). Methods from the HPO thread are structurally incapable of modeling this coupling regardless of iteration budget; methods from the LLM-DBMS thread do not encounter it by construction. Mechanistic ablations confirm that history conditioning and phase-aware transitions are both independently necessary to navigate this coupled space: removing either prevents the agent from locating the compensatory efSearch=64, α=0.80 operating point across all seeds. To our knowledge, this is the first application of LLM-guided optimization to ANN index architecture selection with cross-stage parameter coupling in a vector database management system.

3 THE HICO-DET RETRIEVAL BENCHMARK

3.1 Source Dataset

HICO-DET [4] contains 47,776 images (38,118 train / 9,658 test) annotated across 600 HOI categories with per-instance bounding boxes and interaction labels, where each category is a *verb–object* pair drawn from 117 action verbs and 80 MS-COCO object classes. Three structural properties make HICO-DET distinctively demanding as a benchmark for ANN index configuration optimization. **P1 — Compositional fine-grained discrimination.** The 600 categories span 80 COCO object classes and 117 verb types; categories sharing the same object (e.g., *ride horse*, *sit_on horse*, *straddle horse*) differ only in the action verb. Single-encoder retrieval conflates these closely related categories, making the joint selection of ANN engine and Stage-2 fusion weight critical to performance. **P2 — Long-tail relevance.** Category sizes are highly skewed: the largest categories exceed 1,000 positive images while the smallest have as few as 5. An optimizer that maximizes mean mAP without attending to per-category variance will exploit high-frequency categories while failing on rare but equally weighted queries. **P3 — Dual-encoder modality gap.** CLIP retrieves by language-aligned semantic similarity; DINOv2 captures fine-grained visual structure complementary to CLIP's representations [24]. Neither encoder alone achieves optimal mAP@10, making the Stage-2 fusion weight α a data-dependent parameter that cannot be fixed by hand and must be tuned jointly with the Stage-1 ANN configuration.

Every published use of HICO-DET has been restricted to the detection paradigm [12, 16, 23, 38]: given an image, predict bounding boxes and interaction class labels. We repurpose these rich compositional annotations to construct, to our knowledge, the first text-query image retrieval benchmark from HICO-DET, opening an orthogonal evaluation axis that the detection literature has not explored.

3.2 Benchmark Construction

Retrieval Corpus: Why Merge Train and Test Splits. We use all 47,776 images as a single retrieval corpus. The original HICO-DET train/test split was designed for supervised detection training and carries no meaning for retrieval evaluation, where the task is to rank corpus images given a query, not to generalize across a detection boundary. Merging both splits maximizes the positive-pair density needed for meaningful mAP computation across 600 categories; restricting evaluation to the 9,658-image test split alone would reduce per-category positive counts below the minimum threshold for reliable mAP@10 computation across all 600 classes. CLIP ViT-L/14 [25] embeddings are pre-computed offline using the official OpenAI weights and stored in VDMS as a 768-dimensional L2 DescriptorSet of 47,776 vectors.

Query Design: One Canonical Query per HOI Class. We construct exactly one canonical natural-language text query per HOI category, yielding 600 queries. A single canonical query per category eliminates inter-query variance from paraphrase choice and ensures that each of the 600 HICO-DET categories contributes equally to the mean Average Precision score — a design decision motivated by the same reproducibility principle used in standard IR evaluation [22]. Each query follows the template *"a person {verb} a {object}"*, with the special case *"a person near a {object}"* for the no_interaction verb class. Queries are encoded with the same CLIP ViT-L/14 text encoder, producing 768-dimensional unit-norm text embeddings that reside in the same embedding space as the indexed image vectors.

Ground Truth: Multi-Label Positive Structure. An image is relevant to a query if it carries the corresponding HOI annotation in the HICO-DET ground truth. A single image may be relevant to multiple queries (e.g., an image annotated with both *ride motorcycle* and *hold phone* contributes positives to two queries). Summed over all 600 queries, there are 90,641 positive (image, query) pairs. Every query has at least 5 relevant images, so mAP@10 is defined and non-trivially bounded for each of the 600 queries.

3.3 Evaluation Protocol

We evaluate retrieval quality using mean Average Precision at k=10 (mAP@10), Precision@10, NDCG@10, and Recall@10 over all 600 queries. mAP@10 is our primary metric: it rewards systems that place relevant images in the top-10 positions and is the standard metric for ranked retrieval with multiple relevant documents per query [22]. Throughput is measured in queries per second (QPS), computed as 600 divided by the total wall-clock time for all queries issued as a single batch to a live VDMS container. Batch QPS is the appropriate production metric: a deployed HOI retrieval system serves queries at steady-state rates that are best captured by batch throughput; single-query latency at this scale is dominated by container and network overhead rather than by the ANN index structure, which would mask precisely the index differences the optimizer is designed to exploit. The joint optimization objective is

$$\text{Score} = \text{mAP} \times \text{QPS}. \tag{1}$$

This scalar encodes the accuracy–throughput trade-off symmetrically: gains in QPS and gains in mAP contribute multiplicatively, so neither axis can be sacrificed without penalizing the objective. Section 3.4 below formalizes the configuration space and the optimization problem whose objective is this Score.

3.4 Problem Statement

Definition 1 (Configuration Space)**.** Let Θ denote the joint configuration space of the two-stage pipeline (Section 4). Θ is the Cartesian product of seven parameter domains — *engine*, M/nlist, efSearch/nprobe, k, α, n_{refs}, and ref_strategy — whose individual value domains are enumerated in Section 4. The total cardinality is $|\Theta| = 102{,}144$.

Definition 2 (Evaluation Oracle)**.** For any configuration $\theta \in \Theta$, the evaluation oracle $O(\theta) = (\text{mAP}(\theta), \text{QPS}(\theta))$ is obtained by (i) rebuilding the VDMS ANN index under θ, and (ii) issuing all 600 HICO-DET text queries to the live container and recording mAP@10 (Section 3) and batch QPS. The oracle is *black-box*: no closed-form gradient with respect to θ is available, and each call requires a full index rebuild followed by 600 retrieval operations.

Table 2: Configuration space Θ (Definition 1). †Conditional: valid only for the indicated engine. Cardinalities combine to $|\Theta|$=102,144 (see footnote).

Parameter	Notation	Value Set	Card.
Stage 1 — ANN retrieval			
engine	—	{Flat, HNSWFlat, IVFFlat}	3
graph connectivity† (HNSW)	M	{8, 16, 32, 48, 64}	5
cluster count† (IVF)	nlist	{64, 128, 256, 512}	4
search beam† (HNSW)	efSearch	{32, 64, 100, 150, 200, 300, 500}	7
probed cells† (IVF)	nprobe	{4, 8, 16, 32, 64}	5
candidate pool	k	{50, 100, 150, 200, 300, 500, 750, 1000}	8
Stage 2 — DINOv2 reranking			
fusion weight	α	{0.0, 0.05, 0.10, ..., 0.90}	19
reference count	n_{refs}	{1, 3, 5, 10}	4
ref. strategy	—	{first, centroid, diverse}	3
Total $|\Theta|$ [1]			**102,144**

Definition 3 (Optimization Problem). Given Θ (Definition 1) and O (Definition 2), find

$$\theta^* = \arg\max_{\theta\in\Theta} \text{Score}(\theta), \tag{2}$$

where $\text{Score}(\theta) = \text{mAP}(\theta) \times \text{QPS}(\theta)$ (Eq. (1)).

Intractability. Exhaustive search over $|\Theta| = 102{,}144$ configurations is infeasible under any practical budget: each oracle call requires 3–10 minutes of wall-clock time (index rebuild plus 600 query evaluations on a GPU node), so evaluating the full space would require thousands of GPU-hours. At the N=30-iteration budget used in our study, a non-adaptive strategy covers at most 0.03% of Θ; evaluation feedback is additionally noisy due to VDMS runtime variability. This combination — a black-box, expensive oracle with an exponential search space and a tiny sample budget — is precisely the regime where LLM-guided reasoning offers the greatest advantage over grid and random baselines.

Cross-Stage Parameter Coupling. The seven parameters decompose into two coupled groups: Stage-1 parameters {engine, M, efSearch, nprobe, k} govern ANN recall and throughput, while Stage-2 parameters $\{\alpha, n_{\text{refs}}, \text{ref\_strategy}\}$ govern DINOv2 reranking. The optimal pool size k and fusion weight α are *jointly determined*: at small k, Stage 1 returns only high-confidence CLIP candidates so DINOv2 reranking adds marginally, favouring small α; at large k, the pool contains visually diverse images that benefit from DINOv2 reranking, favouring larger α. This structural coupling invalidates the parameter-independence assumption underlying GP-based Bayesian methods [36] and Optuna TPE [1], which model each parameter's marginal contribution independently and cannot represent cross-stage interactions. Our LLM agent explicitly reasons about these inter-parameter dependencies in its natural-language analysis of prior outcomes (Section 5).

Section Outline. Section 4 constructs the two-stage pipeline that implements oracle O, mapping any $\theta \in \Theta$ to a (mAP, QPS) pair and formally enumerating all parameter domains. Section 5 presents the LLM-guided optimizer that solves $\arg\max_{\theta\in\Theta} \text{Score}(\theta)$ adaptively within N=30 oracle evaluations. Section 6 validates the solution on HICO-DET, GLDv2, and SIFT1M, reporting mean Score across three random seeds against five optimization baselines.

4 TWO-STAGE RETRIEVAL PIPELINE

4.1 Overview

Let $C = \{(\mathrm{c}_i, \mathrm{d}_i)\}_{i=1}^{N}$ denote the corpus of N=47,776 images, where $\mathrm{c}_i \in \mathbb{R}^{768}$ is the unit-normalized CLIP ViT-L/14 embedding and $\mathrm{d}_i \in \mathbb{R}^{1024}$ is the unit-normalized DINOv2 ViT-L/14-reg4 embedding of image i. For HOI class h, let $\mathcal{I}_h \subseteq [N]$ denote the annotated relevant image set. Given a text query q (e.g., *"a person riding a horse"*), the pipeline proceeds in two stages, as illustrated in Figure 2.

Stage 1 encodes q into a 768-dimensional unit-norm query vector $\mathrm{q}_c \in \mathbb{R}^{768}$ using the CLIP ViT-L/14 text encoder and issues an ANN search over the VDMS CLIP DescriptorSet to retrieve a top-k candidate pool $\mathcal{T}_k(q)$. **Stage 2** reranks $\mathcal{T}_k(q)$ using DINOv2 visual distances without requiring any additional index: a per-class reference vector r_h is computed once at initialization from the DINOv2 embeddings of a small representative subset of $\mathcal{I}_h$, and the CLIP and DINOv2 distances of each candidate are fused into a single score. Seven configuration parameters {engine, M or *nlist*, *efSearch* or *nprobe*, k, α, n_{refs}, *ref_strategy*} jointly govern the accuracy–throughput trade-off and are selected by the LLM optimizer described in Section 5.

4.2 Stage 1: CLIP-Based ANN Retrieval

All 47,776 CLIP ViT-L/14 image embeddings are pre-computed offline using the OpenAI weights and stored in VDMS as a 768-dimensional L2 DescriptorSet. At query time, q is encoded by the frozen CLIP text encoder and the resulting q_c is submitted to VDMS as a Find-Descriptor request. VDMS returns the top-k candidate identifiers with their L2 distances $\delta_j^c = \|\mathrm{c}_j - \mathrm{q}_c\|_2$; the candidate pool is formally

$$\mathcal{T}_k(q) = \left\{ j \in [N] : \|\mathrm{c}_j - \mathrm{q}_c\|_2 \le \delta^{(k)} \right\}, \tag{3}$$

where $\delta^{(k)}$ is the k-th smallest L2 distance to q_c across the full corpus. For FaissFlat this set is exact; for FaissHNSWFlat and FaissIVFFlat, VDMS returns a high-recall approximation. Three ANN engines are available, each exposing a different accuracy–throughput operating point:

- **FaissFlat**: exact exhaustive search; highest recall, lowest QPS.
- **FaissHNSWFlat** [21]: graph-based ANN index; accuracy controlled by graph connectivity $M \in \{8, 16, 32, 48, 64\}$ and search beam width $\text{efSearch} \in \{32, 64, 100, 150, 200, 300, 500\}$.
- **FaissIVFFlat** [11]: inverted-file ANN index; accuracy controlled by $\text{nlist} \in \{64, 128, 256, 512\}$ clusters and $\text{nprobe} \in \{4, 8, 16, 32, 64\}$ probed clusters.

Increasing k improves Stage 2 recall at the cost of higher reranking latency; the optimal $k \in \{50, 100, 150, 200, 300, 500, 750, 1000\}$ is jointly selected with the engine parameters.

Figure 2: Two-stage HOI retrieval pipeline. Stage 1: a CLIP ViT-L/14 text embedding drives ANN search in VDMS, returning a top-k candidate pool via a configurable ANN index (FaissFlat, FaissHNSWFlat, or FaissIVFFlat). Stage 2: DINOv2 ViT-L/14-reg4 visual distances rerank the pool on-the-fly via weighted score fusion — no additional index is required at deployment.

4.3 Stage 2: DINOv2 Reranking

Reference vector. DINOv2 ViT-L/14-reg4 is a self-supervised visual encoder that captures fine-grained appearance features — object identity, texture, and spatial structure — complementary to CLIP's language-aligned semantic embeddings. We represent each HOI class h by a reference vector $\mathrm{r}_h \in \mathbb{R}^{1024}$, computed once at initialization as the mean DINOv2 embedding of a selected subset $\mathcal{S}_h \subseteq \mathcal{I}_h$ of $n_r \in \{1, 3, 5, 10\}$ images:

$$\mathrm{r}_h \;=\; \frac{1}{n_r} \sum_{i \in \mathcal{S}_h} \mathrm{d}_i. \tag{4}$$

Three strategies select $\mathcal{S}_h$: **first** takes the first n_r images in dataset order; **centroid** selects the n_r images closest to the class mean $\boldsymbol{\mu}_h^d = \frac{1}{|\mathcal{I}_h|} \sum_{i \in \mathcal{I}_h} \mathrm{d}_i$; **diverse** applies greedy farthest-point sampling, seeding from the most central image and iteratively adding the image furthest from the already-selected set to maximize intra-subset spread. Reference vector construction is a one-time cost (<1 ms per class in numpy) and imposes no per-query overhead.

Distance normalization. CLIP and DINOv2 distances operate at different absolute scales and cannot be directly combined. We apply min-max normalization *within* $\mathcal{T}_k(q)$ to map both onto [0, 1]:

$$\hat{c}_j = \frac{\delta_j^c - \min_{i \in \mathcal{T}_k} \delta_i^c}{\max_{i \in \mathcal{T}_k} \delta_i^c - \min_{i \in \mathcal{T}_k} \delta_i^c + \varepsilon}, \tag{5}$$

$$\hat{d}_j = \frac{\delta_j^d - \min_{i \in \mathcal{T}_k} \delta_i^d}{\max_{i \in \mathcal{T}_k} \delta_i^d - \min_{i \in \mathcal{T}_k} \delta_i^d + \varepsilon}, \tag{6}$$

where $\delta_j^d = \|\mathrm{d}_j - \mathrm{r}_h\|_2$ is the DINOv2 distance of candidate j to the class reference vector and $\varepsilon = 10^{-10}$ prevents division by zero. Normalizing within the candidate pool — rather than globally across the corpus — ensures the two scores remain commensurate regardless of the per-query scale of retrieved distances.

Fusion and final ranking. The fusion score for candidate $j \in \mathcal{T}_k(q)$ is the convex combination

$$s_j \;=\; (1 - \alpha)\, \hat{c}_j \;+\; \alpha\, \hat{d}_j, \tag{7}$$

where $\alpha \in \{0.0, 0.05, \ldots, 0.90\}$ governs the DINOv2 contribution. Lower s_j indicates higher relevance. The final ranked list is $\mathcal{T}_k(q)$ sorted by s_j in ascending order. Setting α=0 reduces the system to CLIP-only ranking; α=1 reduces it to DINOv2-only ranking. The optimal α is data-dependent and is selected jointly with the ANN engine configuration by the optimizer in Section 5.

4.4 Why Text Queries Outperform Image Queries: Centroid Geometry

We formalize why a text query systematically outperforms a single reference image on compositionally structured HOI categories, grounding the argument in the geometry of CLIP's joint embedding space.

Let $\boldsymbol{\mu}_h^c = \frac{1}{|\mathcal{I}_h|} \sum_{i \in \mathcal{I}_h} \mathrm{c}_i$ denote the visual centroid of class h in CLIP space, and let $\sigma_h^2 = \frac{1}{|\mathcal{I}_h|} \sum_{i \in \mathcal{I}_h} \|\mathrm{c}_i - \boldsymbol{\mu}_h^c\|_2^2$ be the intra-class variance. For any query vector $\mathrm{q}' \in \mathbb{R}^{768}$, the bias–variance decomposition of the expected squared L2 distance to a uniformly sampled class member c_i ($i \sim \mathcal{I}_h$) is:

$$\mathbb{E}_i\big[\|\mathrm{q}' - \mathrm{c}_i\|_2^2\big] \;=\; \underbrace{\|\mathrm{q}' - \boldsymbol{\mu}_h^c\|_2^2}_{\text{bias}^2} \;+\; \underbrace{\sigma_h^2}_{\text{variance}}. \tag{8}$$

The variance term σ_h^2 is fixed by the data; only the bias term $\|\mathrm{q}' - \boldsymbol{\mu}_h^c\|_2^2$ depends on the choice of query.

CLIP's contrastive training aligns the text embedding of a category description with the visual embeddings of matching images [25]. For a compositionally structured category such as *"a person riding a horse"* — which spans diverse viewpoints, backgrounds, and subjects — the text embedding aggregates signals from many matched images and therefore lies near the visual centroid of the class: $\mathrm{q}_c \approx \boldsymbol{\mu}_h^c$, making the bias term in Eq. (8) approximately zero:

$$\mathbb{E}_i\big[\|\mathrm{q}_c - \mathrm{c}_i\|_2^2\big] \;\approx\; \sigma_h^2. \tag{9}$$

In contrast, a reference image c_j ($j \in \mathcal{I}_h$) lies at displacement $\varepsilon_j = \|\mathrm{c}_j - \boldsymbol{\mu}_h^c\|_2 > 0$ from the centroid, contributing a positive bias term:

$$\mathbb{E}_i\big[\|\mathrm{c}_j - \mathrm{c}_i\|_2^2\big] \;=\; \varepsilon_j^2 + \sigma_h^2 \;>\; \sigma_h^2. \tag{10}$$

The text query therefore achieves lower expected squared distance to every class member than any individual reference image, by a margin of ε_j^2. This advantage is largest for visually diverse categories: for an HOI class spanning many viewpoints and subjects (e.g., *ride horse*), any single reference image lies far from the centroid (ε_j large) and provides a poor proxy for the full category distribution, amplifying the margin over a text query. Under L2-based ANN retrieval, lower expected distance to class members translates directly to higher recall at any fixed k, and therefore to higher mAP.

5 LLM-GUIDED ANN HYPERPARAMETER OPTIMIZATION

Figure 3 illustrates the overall optimization framework. Five methods — LLM Agent, Optuna TPE, GP-BO, Grid Search, and Random Search — each propose a joint index configuration (engine, M, *efSearch*/*nlist*/*nprobe*, pool size k, fusion weight α, reference count n_r, reference strategy) over $N = 30$ iterations. Each proposed configuration is loaded into a VDMS Apptainer container, which builds the ANN index and executes all 600 HICO-DET queries; the resulting mAP and QPS are combined into Score = mAP × QPS (HICO-DET and GLDv2). Adaptive methods — LLM Agent, Optuna TPE, and GP-BO — receive the score signal as feedback; Grid Search and Random Search are non-adaptive baselines that do not condition on observed outcomes. GP-BO uses optuna.samplers.GPSampler (Gaussian Process surrogate with Expected Improvement), directly replicating VDTuner's [36] parameter-independent GP paradigm adapted to our single-objective Score.

5.1 Agent Design

The LLM agent operates as a closed-loop optimizer over a mixed-type search space of approximately 102,000 configurations. Engine-conditional parameters are $M \in \{8, 16, 32, 48, 64\}$ and efSearch $\in \{32, 64, 100, 150, 200, 300, 500\}$ for FaissHNSWFlat, and nlist $\in \{64, 128, 256, 512\}$ with nprobe $\in \{4, 8, 16, 32, 64\}$ for FaissIVFFlat. Shared parameters are pool size $k \in \{50, \ldots, 1000\}$, fusion weight $\alpha \in \{0.0, 0.05, \ldots, 0.90\}$, reference count $n_r \in \{1, 3, 5, 10\}$, and reference strategy $\in$ {*first*, *centroid*, *diverse*}.

Prompt structure. At each iteration t, the agent receives a single prompt containing: (i) a description of the VDMS two-stage pipeline architecture and per-engine parameter physics — including the known dependency that efSearch and M must be co-tuned, and that *efSearch* $\ll k$ accepts ANN approximation error in exchange for throughput; (ii) a *per-iteration diagnostic* derived from the observed Score relative to the running best, selecting one of five feedback classes (*excellent*, *explore_more*, *low_mAP*, *low_QPS*, or *moderate*) and providing targeted guidance on which parameter axis to adjust; (iii) the complete ordered history of all $t-1$ evaluated (configuration, Score, mAP, QPS) tuples; (iv) a dynamically computed list of untried α values and untried (n_r, ref_strategy) pairs, derived at inference time from the history; and (v) seven decision rules governing structural diversity, stagnation escape, and an anti-collapse constraint that prohibits consecutive iterations that vary only α without any structural change. The agent responds with a structured JSON object specifying the next configuration and a *reasoning* field articulating its rationale. Figure 4 illustrates the prompt template.

Phase-aware strategy. The N-iteration budget is partitioned into three phases whose boundaries scale with N: **Exploration** (iterations 1–$\lfloor 0.4N \rfloor$) broadly varies engine, M, and k across structurally distinct regions to map the Score landscape; **Exploitation** (iterations $\lfloor 0.4N \rfloor$+1–$\lfloor 0.75N \rfloor$) refines the most promising engine–M combination found during exploration, focusing on α, n_r, and reference strategy; and **Fine-tuning** (iterations $\lfloor 0.75N \rfloor$+1–N) performs narrow coordinate refinement of α and reference strategy around the current best. This schedule encodes the domain insight that structural choices (engine, M) determine the achievable QPS ceiling and must be resolved before continuous choices (α, n_r) are optimized — a conditional dependency between parameters that Optuna TPE, which models each parameter independently via univariate kernel density estimation, cannot exploit.

Duplicate avoidance. Each proposal is checked against the complete evaluation history. If a duplicate configuration is detected, the prompt is resubmitted with an explicit rejection notice listing the already-evaluated configuration and its score. Up to four retries are attempted; after four consecutive duplicates, a uniformly random valid configuration is selected as a fallback, ensuring the optimization loop continues without stalling.

6 EXPERIMENTS

6.1 Experimental Setup

Datasets. We evaluate on three benchmarks. **HICO-DET** (Section 3): 47,776 images, 600 compositional text queries, 90,641 ground-truth positive pairs — a stringent testbed combining fine-grained HOI categories, long-tailed relevance distributions, and a cross-modal (text-to-image) retrieval setting. **GLDv2** [34]: Google Landmarks Dataset v2, a large-scale image-to-image retrieval benchmark with 762,000 gallery images and 1,129 public queries spanning global landmarks, providing a 16× larger corpus for cross-domain generalization evaluation. **SIFT1M** [10]: a standard ANN benchmark of one million 128-dimensional SIFT feature vectors with 10,000 queries and exact ground-truth 10-nearest neighbors, providing a complexity-matched control — the index space reduces to a single-stage HNSW problem with no cross-stage coupling — to test whether the LLM's advantage is specific to coupled search spaces. For HICO-DET, image embeddings are 768-dimensional CLIP ViT-L/14 vectors. For GLDv2, each image is represented by a 768-dimensional CLIP ViT-L/14 embedding and a 1024-dimensional DINOv2 ViT-L/14-reg4 embedding, both stored in VDMS. For SIFT1M, raw 128-dimensional SIFT vectors are stored directly in VDMS without any reranking stage; the optimization objective is Score = Recall@10× QPS.

Compared Methods. We compare four optimization strategies, each run for N=30 iterations:

- **LLM Agent (Ours):** at each iteration, the agent receives the full history of (configuration, Score) pairs in its prompt and proposes the next configuration; implemented with MiniMax-M2.1 via OpenRouter.
- **Optuna TPE** [1]: Tree-structured Parzen Estimator Bayesian optimization; adaptive, conditions on observed outcomes (3 seeds: 42, 99, 200).
- **Random Search** [3]: samples configurations uniformly at random; non-adaptive (3 seeds: 78, 99, 200).
- **Grid Search**: enumerates a fixed grid of configurations in lexicographic order; non-adaptive.

As system baselines we include UniIR [33] (CLIP ViT-L/14, brute-force), standalone CLIP ViT-L/14 (FaissFlat), standalone DINOv2 ViT-L/14-reg4 (FaissFlat), and CLIP+DINOv2 reranking (FaissFlat), all using exact exhaustive search to establish accuracy upper bounds independent of ANN approximation.

Metrics. Retrieval quality is measured by mean Average Precision at k=10 (mAP@10), the standard metric for ranked retrieval with multiple relevant documents [22], as well as Precision@10, NDCG@10, and Recall@10. Throughput is measured in queries per second (QPS), averaged over all 600 queries issued as a single batch to the VDMS container. The joint optimization objective is Score = mAP × QPS, which rewards configurations that simultaneously achieve high retrieval quality and high throughput, and penalizes any single-axis extreme equally.

Implementation. All experiments run on the NCSA Delta GPU cluster (partition gpuA40x4): one NVIDIA A40 GPU (48 GB VRAM), 16 CPU cores (AMD EPYC 7763), and exclusive node access. VDMS is deployed as an Apptainer container (v2.2.1); all retrieval and optimization code is implemented in Python 3.12. For Optuna, we

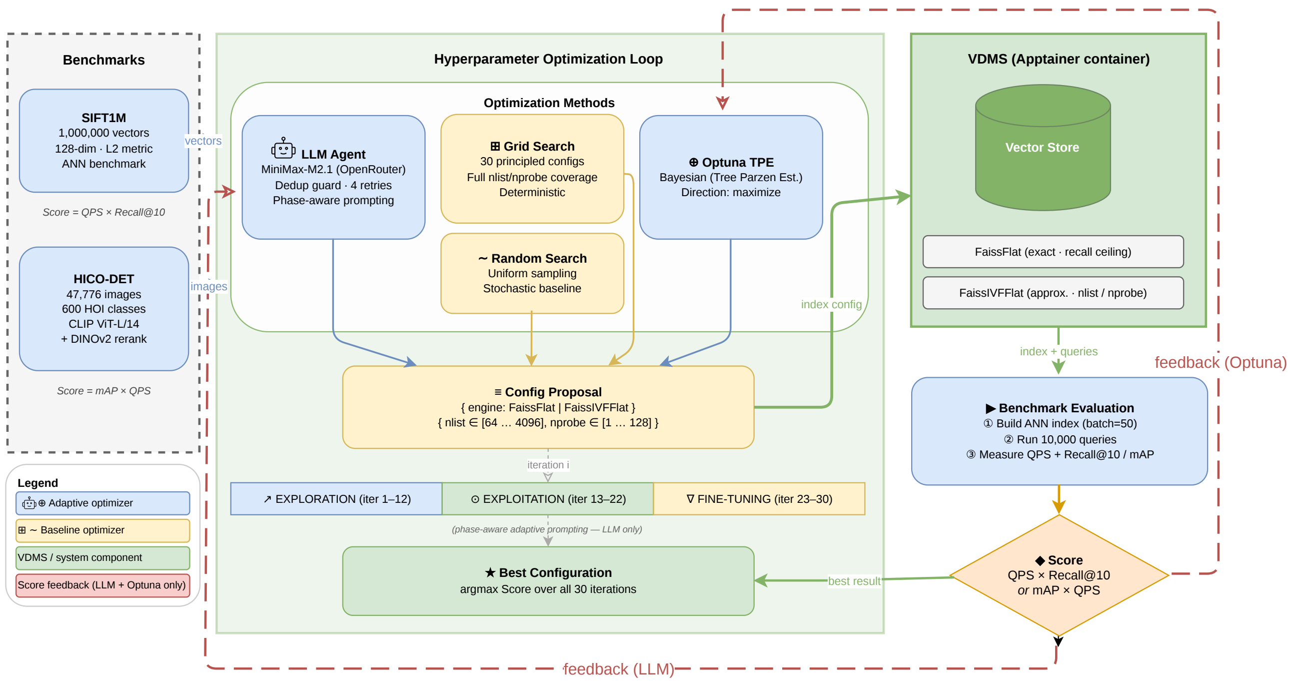

Figure 3: Overview of the Agentic VDMS optimization framework. Five methods — LLM Agent (phase-aware, adaptive), Optuna TPE (Tree Parzen Estimator, adaptive), and GP-BO (Gaussian Process surrogate, adaptive) vs. Grid Search and Random Search (non-adaptive baselines) — each propose ANN index configurations evaluated by VDMS over N=30 iterations. Score = mAP × QPS (HICO-DET and GLDv2). Red dashed arrows indicate score feedback; only adaptive methods receive it.

use the official optuna library [1] with the default TPE sampler. Each optimization iteration builds a complete ANN index from scratch and issues all 600 queries in a single batch, requiring approximately 7–8 minutes of wall-clock time. The full 30-iteration budget therefore demands approximately 3.5–4 hours of compute per method, making sample efficiency — how quickly each method reaches a high-scoring configuration — a practically meaningful criterion. We fix N=30 for all methods and datasets for three reasons. First, 30 iterations represent under 1% coverage of the valid configuration space (e.g., 5,940 HNSW configs on SIFT1M), placing all methods in a regime where optimizer choice is genuinely consequential rather than trivially solvable by exhaustive enumeration. Second, 3.5–4 hours of wall-clock tuning time per method is a realistic practitioner budget; extending to N=100 would require over 12 hours per method without changing the ranking, since our convergence analysis (Figure 5) shows the LLM agent's advantage is established by iteration 10 and the gap only widens thereafter. Third, N=30 is consistent with the evaluation protocol of closely related LLM-guided and Bayesian optimization studies [13, 35].

6.2 System Baselines

Table 3 reports exact-search system baselines that isolate retrieval accuracy from ANN approximation error, establishing upper bounds for each modeling choice.

Compositional difficulty. Even with exhaustive nearest-neighbor search and DINOv2 reranking, the best achievable mAP is only 0.276 (CLIP+DINOv2, FaissFlat, k=2000). This ceiling reflects the structural difficulty of fine-grained HOI categories: classes such as *ride horse*, *sit_on horse*, and *straddle horse* share the same object but differ only in the action verb, and CLIP's joint embedding space partially conflates these distinctions.

UniIR comparison. To provide a fair, apples-to-apples comparison, we evaluate all four UniIR variants (CLIP-SF and BLIP-FF, image-only and multimodal) through the same VDMS FaissFlat index using the identical sequential client-server QPS measurement protocol. UniIR's fine-tuned CLIP-SF achieves mAP = 0.095 in VDMS — a 2.6× drop from the standard CLIP ViT-L/14 zero-shot retrieval (mAP = 0.246, brute-force). This degradation arises because UniIR's multi-task fine-tuning on COCO-style image-text pairs disrupts the precise HOI semantic alignment that the original CLIP text encoder provides for compositional queries such as *"a person ride a motorcycle"*. The BLIP-FF variants perform similarly (mAP ≈ 0.089), confirming that the bottleneck is task-specific semantic alignment rather than model architecture. Despite higher QPS (≈19.7) due to their smaller k=10 search, all UniIR variants achieve Score ≤ 1.871, below our unoptimized CLIP-only baseline (Score = 1.707 at k=500, QPS = 8.21) once mAP differences are accounted for, and far below the LLM-optimized result reported in Section ??.

Modality gap. DINOv2 alone (α=1.0, k=2000, mAP = 0.217) underperforms CLIP alone at the same k (mAP = 0.240) by 2.3 percentage points, confirming that language-aligned text embeddings

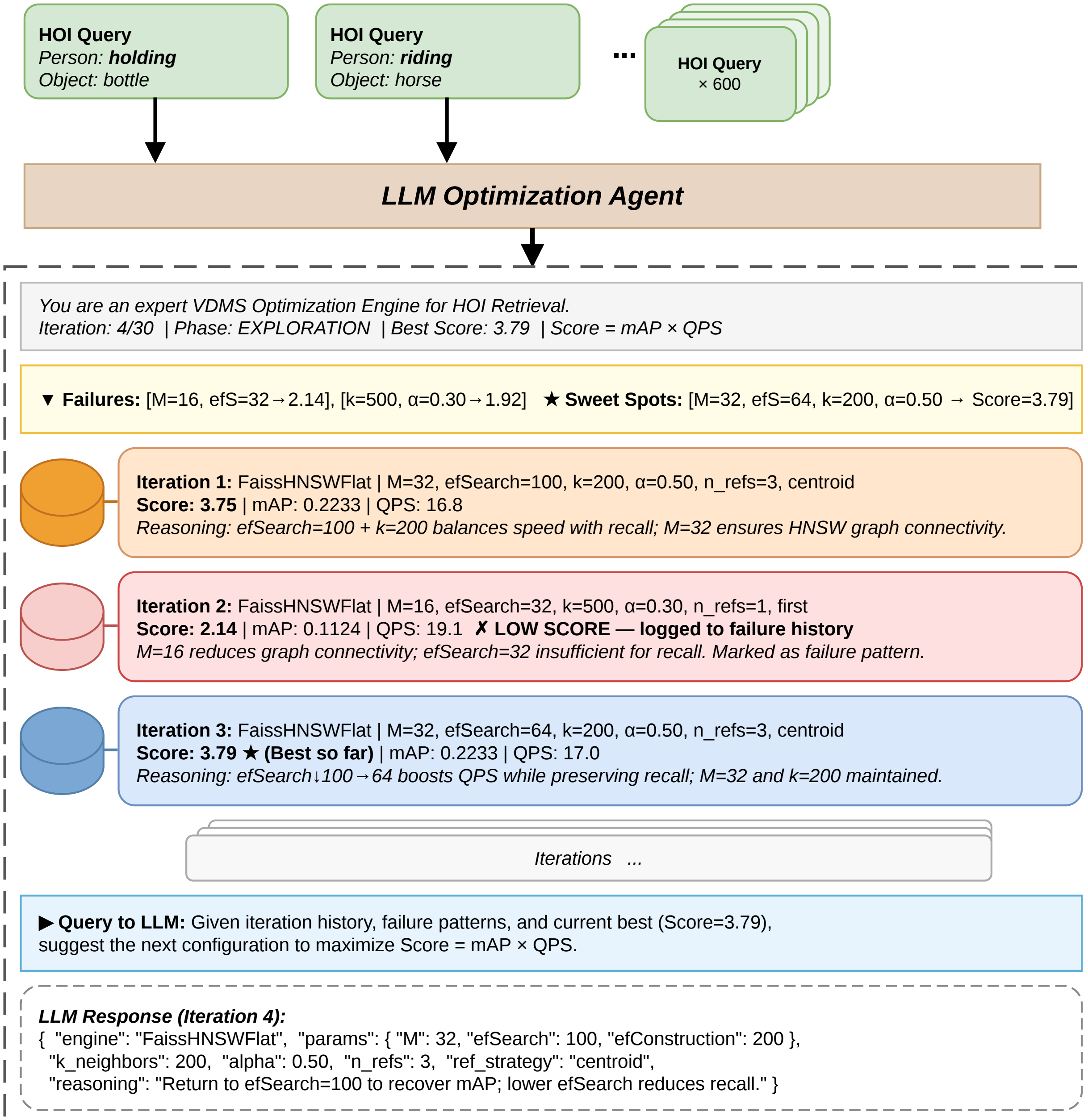

Figure 4: LLM prompt structure used by the agent for ANN index hyperparameter optimization. The system prompt establishes the optimization objective; the user prompt supplies the measured Score history and asks the agent to propose the next configuration.

are essential for compositional HOI retrieval. Visual similarity alone fails to discriminate action-level distinctions between HOI categories that share the same object. However, the two modalities are complementary rather than competing: weighted fusion at α=0.5 (k=2000) achieves mAP = 0.276, a gain of +3.65 pp over CLIP alone (+15.2% relative) at negligible latency cost — DINOv2 reranking adds only 4.82 ms over an 855 ms CLIP retrieval pass, a 0.6% overhead. This complementarity motivates including α as a primary optimization target and validates the two-stage pipeline design.

Table 3: System baselines on HICO-DET (47,776 images, FaissFlat exact search). Score = mAP × QPS. α is the DINOv2 reranking weight (α=0: CLIP only; α=1: DINOv2 only); "—" indicates not applicable. UniIR variants are measured in VDMS under identical conditions (FaissFlat, sequential client-server QPS); UniIR zero-shot is a brute-force numpy reference with no database QPS. System efficiency (latency, build time) is reported in Table 5.

Method	α	k	mAP	P@10	R@10	QPS	Score
UniIR [33] baselines							
Zero-shot CLIP ViT-L/14 (brute-force)	—	—	0.2461	0.3370	0.0967	—	—
CLIP-SF Large, image-only (VDMS)	—	10	0.0951	0.4477	0.1327	19.67	1.871
CLIP-SF Large, multimodal (VDMS)	—	10	0.0894	0.4415	0.1206	19.46	1.740
BLIP-FF Large, image-only (VDMS)	—	10	0.0889	0.4420	0.1213	19.69	1.751
BLIP-FF Large, multimodal (VDMS)	—	10	0.0745	0.4097	0.1019	19.72	1.469
Our CLIP + DINOv2 pipeline (FaissFlat exact)							
CLIP ViT-L/14	0.0	500	0.2079	0.3370	0.0967	8.21	1.707
CLIP ViT-L/14	0.0	2000	0.2398	0.3370	0.0967	1.17	0.280
DINOv2 ViT-L/14-reg4	1.0	2000	0.2165	0.3232	0.0800	1.16	0.250
CLIP + DINOv2 rerank	0.5	2000	**0.2763**	**0.3812**	**0.1138**	1.15	0.319

Accuracy–throughput trade-off. Reducing k from 2000 to 500 in CLIP-only retrieval (FaissFlat, α=0) raises QPS by 7.0× (8.21 vs. 1.17) at the cost of only 3.2 percentage points of mAP (0.208 vs. 0.240). The disproportionate Score gain (1.707 vs. 0.280, a 6.1× increase) illustrates a fundamental asymmetry in the Score objective: throughput gains in the high-QPS regime yield multiplicatively larger Score improvements than equivalent mAP gains, because the derivative $\partial\text{Score}/\partial\text{QPS}$ = mAP exceeds $\partial\text{Score}/\partial\text{mAP}$ = QPS whenever QPS $\gg$ mAP. This asymmetry is precisely the non-linear trade-off that Score = mAP × QPS encodes, and it cannot be navigated effectively without conditioning each new proposal on the full history of observed outcomes.

6.3 Optimizer Comparison and Sample Efficiency

Overall performance. Table 4 and Figure 5 report the results of all four optimization methods over 30 iterations on HICO-DET. The LLM agent achieves the highest mean Score = 4.44, a 2.5× improvement over the CLIP FaissFlat baseline at k=500 (Score = 1.707) and 13.9× over brute-force CLIP+DINOv2 reranking (Score = 0.319). At its optimal configuration — FaissHNSWFlat with M=48, efSearch=64, k=200, α=0.80, n_r=10, and *centroid* reference selection — the pipeline achieves mAP = 0.255 and QPS = 16.98, representing a 14.7× throughput gain over brute-force CLIP+DINOv2 (QPS = 1.15) at a cost of only 2.1 percentage points of mAP (0.255 vs. 0.276). Optuna TPE (mean Score = 4.05) and Random Search (mean Score = 4.04) reach comparable final scores, trailing the LLM agent by 8.8% and 9.0% respectively. Grid Search (mean Score = 3.44) performs substantially worse, running all 30 iterations but plateauing at iteration 19 and achieving mAP only 0.169 — below the CLIP-only exact-search baseline at k=2000 (mAP = 0.240).

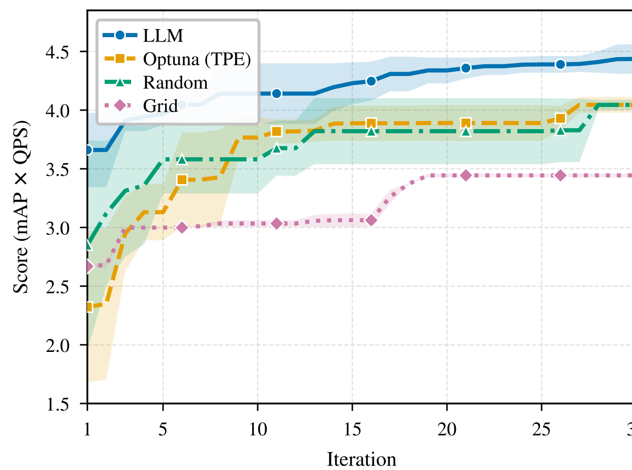

Figure 5: Convergence of optimization methods on HICO-DET over 30 iterations (mean ±1σ across 3 seeds; 2 runs for Grid). Best Score (mAP × QPS) achieved so far is plotted at each iteration. The LLM agent reaches Score ≈ 3.97 by iteration 6 — a level that Optuna TPE first matches at iteration 27 and Random Search first matches at iteration 28 — and continues improving to Score = 4.44 at iteration 30. Grid Search plateaus at iteration 19, having exhausted its fixed 30-configuration sweep.

Sample efficiency. The LLM agent's primary advantage lies in *sample efficiency*: its mean convergence curve reaches Score ≈ 3.97 by iteration 6, already exceeding Grid Search's final best (mean Score = 3.44) and surpassing both Optuna TPE (mean Score = 3.41 at iteration 6) and Random Search (mean Score = 3.58 at iteration 6). Optuna TPE first exceeds the LLM agent's iteration-6

Table 4: Comparison of ANN index hyperparameter optimization methods on HICO-DET (30 iterations, best config reported). Score = mAP × QPS. *Iters→Best*: first iteration achieving the reported best score. † Grid Search ran all 30 iterations, plateauing at iteration 19. All metrics are means across 3 seeds (LLM, Optuna, Random); Grid is the mean of 2 runs. The LLM agent achieves the highest Score and reaches Score ≈ 3.97 within 6 iterations — a level requiring 27 iterations for Optuna TPE and 28 for Random Search. ‡ P@10 was not recorded for Grid Search runs. § QDS uses a single run with query difficulty-weighted objective (Section 6.6); Score and mAP are evaluated on the standard unweighted metric for fair comparison. Full per-query latency breakdown is reported in Table 5.

Method	Score↑	mAP↑	P@10↑	QPS↑	Iters→Best
Grid Search†‡	3.44	0.169	—	20.42	19/30
Random Search	4.04	0.260	0.569	16.13	28/30
Optuna TPE	4.05	0.256	0.529	15.84	27/30
LLM Agent (Ours)	**4.44**	0.259	0.487	17.11	30/30
LLM Agent + QDS (Ours)§	**4.66**	0.243	—	**19.17**	19/30

Table 5: System efficiency for representative configurations on HICO-DET (47K images, 600 queries, averaged over 5 runs). *Build(s)*: one-time index construction time. *CLIP(ms), DINO(ms)*: per-query stage latencies. *Lat(ms)*: total end-to-end latency per query. *QPS*: queries per second. HNSW builds are slower than Flat due to graph construction but retrieve far faster.

Configuration	Engine	k	Build(s)↓	CLIP(ms)↓	DINO(ms)↓	Lat(ms)↓	QPS↑
Exact-search baselines							
CLIP only (α=0)	FaissFlat	500	433.3	121.0	0.03	121.1	8.21
CLIP only (α=0)	FaissFlat	2000	431.2	850.0	0.09	850.1	1.17
DINOv2 only (α=1)	FaissFlat	2000	432.9	851.6	5.18	856.8	1.16
CLIP+DINOv2 (α=0.5)	FaissFlat	2000	430.1	855.2	4.82	860.0	1.15
Optimizer best configurations							
LLM Agent (Ours)	HNSW M=48	200	441.8	58.2	0.49	58.9	16.98
Optuna TPE	HNSW M=32	200	423.3	62.0	0.47	62.6	15.97
Random Search	HNSW M=32	200	444.3	48.3	0.32	48.9	20.47
Grid Search	HNSW	—	~445	—	—	—	20.51

threshold at iteration 27; Random Search does not match it until iteration 28. At ≈7–8 minutes of wall-clock time per evaluation, this translates to concrete practical savings: the LLM agent reaches Score ≥ 3.97 approximately 2.5 hours ahead of Optuna TPE (22 fewer evaluations) and over 2.5 hours ahead of Random Search (23 fewer evaluations). This efficiency stems directly from the agent's ability to condition each new proposal on the full empirical history and its phase-aware schedule that resolves structural choices (engine, M) before continuous choices (α, n_r) — a conditional dependency structure that neither independent density estimators nor blind sampling can exploit.

Configuration insight. The LLM agent's optimal configuration reveals a non-obvious operating point that would be unlikely to emerge from unguided search within 30 iterations. The agent selects efSearch=64, well below the conventional HNSW heuristic of efSearch ≥ k [21], knowingly accepting higher ANN approximation error in exchange for a throughput advantage. This deliberate Stage-1 recall sacrifice is compensated at Stage 2 by raising α to 0.80, placing 80% of the final score weight on DINOv2 visual similarity to recover relevant images that approximate CLIP ranking may have missed. The agent further discovers that n_r=10 centroid-selected reference images substantially outperform n_r=1: averaging ten images closest to the DINOv2 class centroid produces a stable reference vector that minimizes the bias term ε_j^2 in Eq. (10), whereas any single image is displaced from the mean by a non-zero amount that degrades reranking precision. The three-way interaction among efSearch, k, and α is the joint optimum region that non-adaptive and parameter-independent methods systematically fail to locate within a 30-iteration budget.

Failure modes of non-adaptive methods. Grid Search's poor performance is a direct consequence of its one-dimensional sweep structure: it sweeps k while holding M=32 and α=0.30 fixed, never testing M=48 (the LLM's optimal) and never exploring $\alpha > 0.30$ (vs. the optimal α=0.80). The lexicographic enumeration order guarantees that the joint optimum of (M, k, α, n_r, ref_strategy) cannot be reached within a 30-iteration budget. Optuna TPE exhibits premature convergence: at iterations 11–13, all three consecutive evaluations produce configurations with identical mAP = 0.169 and QPS = 18.6, a plateau that reflects the structural limitation of TPE's univariate kernel density estimator on a search space with hard conditional dependencies between engine type and its valid parameters. The TPE sampler, lacking any representation of the conditional relationship between engine and its parameters, repeatedly revisits a low-scoring FaissIVFFlat region before eventually locating the global optimum at iteration 27 — four iterations after the LLM agent has already committed to and begun refining its HNSW-based operating point.

Table 6: Optimizer comparison on GLDv2 (762K gallery, 1,129 queries, 30 iterations, seed 42). Score = mAP × QPS. *Iters→Best*: first iteration achieving the reported best score.

Method	Score↑	mAP↑	P@10↑	QPS↑	Iters→Best
Optuna TPE	3.6436	0.1785	0.2128	20.41	25/30
Random Search	3.6519	0.1784	0.2131	20.47	8/30
Grid Search	3.6862	0.1788	0.2132	20.62	7/30
LLM Agent (Ours)	**3.7093**	**0.1788**	**0.2131**	**20.75**	**1/30**

Table 7: Optimizer comparison on SIFT1M (1M SIFT vectors, 10,000 queries, 30 iterations, seed 42). Score = Recall@10 × QPS. *Iters→Best*: first iteration achieving the reported best score.

Method	Score↑	Recall@10↑	QPS↑	Iters→Best
Random Search	1014.2	0.912	1111.6	23/30
Optuna TPE	1021.1	0.888	1150.0	14/30
Grid Search	1035.0	0.952	1087.1	25/30
LLM Agent (Ours)	**1036.0**	**0.941**	**1100.8**	**17/30**

6.4 Cross-Domain Generalization: GLDv2 Landmark Retrieval

To test whether the LLM agent's advantage generalizes beyond the HICO-DET benchmark, we evaluate all four optimization methods on the Google Landmarks Dataset v2 (GLDv2) [34], a large-scale image-to-image retrieval benchmark with 762,000 gallery images and 1,129 public queries spanning global landmarks. GLDv2 differs from HICO-DET in two key respects: the corpus is 16× larger (762K vs. 47K images), stressing ANN index construction time and memory, and the retrieval task is image-to-image rather than text-to-image, removing the text-query centroid advantage described in Section 4. Each gallery image is represented by a 768-dimensional CLIP ViT-L/14 embedding and a 1024-dimensional DINOv2 ViT-L/14-reg4 embedding stored in VDMS; the same Score = mAP × QPS objective and 30-iteration budget are used.

Overall performance. Table 6 reports the GLDv2 results. The LLM agent achieves Score = **3.7093**, the best result across all methods, outperforming Grid Search (Score = 3.6862), Random Search (Score = 3.6519), and Optuna TPE (Score = 3.6436). The pattern mirrors HICO-DET: the LLM agent ranks first on both datasets despite the 16× scale increase and the shift from text-to-image to image-to-image retrieval, demonstrating cross-domain generalization of the LLM-guided optimization approach.

Scale effects. At 762K images, each optimization iteration requires approximately 5–6 minutes of wall-clock time, roughly 2× longer than HICO-DET, due to HNSW index construction at larger scale (≈265 s build time vs. ≈90 s). All 30-iteration budgets therefore complete in ≈3 hours per method.

Sample efficiency. The LLM agent finds its best configuration at iteration 1 — identifying the optimal k=100, α=0.90, efC=400, centroid-PRF setting immediately from its domain knowledge about index quality and retrieval trade-offs. Grid Search requires 7 evaluations to reach a comparable config, Random Search 8, and Optuna TPE 25. This extreme sample efficiency (6× fewer evaluations than Grid Search, 25× fewer than Optuna) is the clearest demonstration that LLM guidance substitutes for exhaustive search even at large scale.

6.5 Complexity-Matched Validation: SIFT1M

To test whether the LLM agent's advantage is specific to search spaces with cross-stage coupling, we evaluate four methods (LLM Agent, Optuna TPE, Grid Search, Random Search) on SIFT1M [10] under the same 30-iteration budget; GP-BO is omitted from this ablation as it targets the coupled-space comparison on HICO-DET and GLDv2. Unlike HICO-DET, SIFT1M involves no reranking stage and no fusion weight α: the optimizer tunes only HNSW graph connectivity M, construction depth *efConstruction*, search depth *efSearch*, and candidate pool size k — four parameters with no cross-stage coupling. This reduced, structurally simpler space admits no cross-stage dependencies by construction, providing a clean complexity-matched control.

Overall performance. Table 7 reports the results. All four methods converge to FaissHNSWFlat with *efSearch*$\in \{24, 32\}$ as the optimal operating point, yielding final Scores of 1036.0 (LLM), 1035.0 (Grid), 1021.1 (Optuna), and 1014.2 (Random). The spread across all four methods is only 2.1% — a stark contrast to HICO-DET, where the LLM leads by 9.7% over Optuna TPE, 10.4% over Random, and 3.3% over GP-BO.

Complexity-proportional advantage. The result confirms the paper's central theoretical prediction: the LLM agent's advantage is complexity-proportional. On HICO-DET, the 7-parameter coupled space (efSearch–k–α interaction across stages) rewards history-conditioned architectural reasoning; removing the cross-stage coupling, as on SIFT1M, eliminates the advantage because the residual 4-parameter ANN search space (engine, M, efSearch, k) is shallow enough for random and grid search to locate the optimum within the same 30-iteration budget. This pattern is consistent with the abstract's theoretical prediction that the agent's benefit scales with the coupling complexity of the configuration landscape.

Sample efficiency. The LLM agent reaches Score ≥ 1000 from iteration 1, while Grid Search requires 4 iterations, Optuna TPE requires 11, and Random Search requires 23. On this simpler benchmark, however, the practical gap is smaller: all methods finish within the 30-iteration budget at roughly comparable final performance, and none exhibits the premature convergence plateau seen in HICO-DET's Optuna runs.

6.6 Ablation Study

We isolate the contribution of individual pipeline decisions by comparing performance across systematically varied configurations, using exact-search baselines (Table 3) to control for ANN approximation.

DINOv2 reranking weight (α). Table 3 provides a controlled three-point ablation of α under FaissFlat exact search at k=2000, where approximation error does not confound the comparison. At α=0.0 (CLIP alone), mAP = 0.240. At α=0.5, mAP rises to 0.276 (+3.65 pp, +15.2% relative), with QPS essentially unchanged at 1.15 vs. 1.17 — a near-free accuracy improvement. At α=1.0 (DINOv2 alone), mAP falls to 0.217, confirming that the visual encoder cannot resolve action-level distinctions without language-aligned Stage-1 retrieval. Together, these three points establish that $\alpha \in (0, 1)$ strictly dominates both pure-CLIP and pure-DINOv2 retrieval, and that the optimal α under exact search lies near 0.5. In the ANN setting, the LLM agent discovers α=0.80 — substantially higher than the exact-search optimum — a shift consistent with the reranker serving as an *error corrector* for Stage-1 approximation: at efSearch=64 $<$ k=200, HNSW necessarily misses some relevant neighbors, and increasing α compensates by weighting fine-grained visual reranking more heavily over the returned pool.

Candidate pool size (k). Pool size k controls how many items the DINOv2 reranker sees, trading Stage-1 recall coverage against Stage-2 reranking latency. From Table 3, increasing k from 500 to 2000 at α=0 (FaissFlat) improves mAP from 0.208 to 0.240 (+3.2 pp) but reduces QPS from 8.21 to 1.17 (7.0× throughput loss), yielding a net Score reduction from 1.707 to 0.280. The Score objective therefore strongly favors smaller k: each additional candidate examined adds roughly equal latency whether or not it is relevant, while the mAP gains from a larger pool are subject to diminishing returns as k already covers the majority of relevant items. The LLM agent selects k=200 — smaller than any of the exact-search baseline values — on the reasoning that a larger reference set (n_r=10) and higher reranking weight (α=0.80) compensate for the narrower initial pool, maintaining mAP = 0.255 while preserving the throughput advantage that Score rewards.

Reference selection strategy and count (n_r, ref_strategy). The DINOv2 reranker constructs a class reference vector from n_r corpus images via one of three strategies: *first* (top-1 CLIP-ranked image), *centroid* (mean DINOv2 embedding of the n_r images closest to the class centroid), and *diverse* (greedy farthest-point selection for maximum within-class spread). The LLM agent converges to *centroid* with n_r=10, consistent with the geometric analysis in Section 4: the centroid strategy explicitly minimizes the expected displacement ε_j between the reference vector and the HOI class mean, reducing the reranking bias term ε_j^2 in Eq. (10). The *first* strategy depends on a single CLIP-ranked image that may lie arbitrarily far from the class mean — particularly harmful for the long-tailed categories that dominate HICO-DET. The *diverse* strategy maximizes spread at the cost of mean proximity, yielding a reference vector that spans the class but does not minimize bias. Averaging n_r=10 centroid-proximal images produces a reference that is simultaneously low-bias and low-variance, making it the most reliable proxy for the category distribution.

ANN engine selection. Both the LLM agent and Optuna TPE converge to FaissHNSWFlat as the optimal engine. HNSW's graph-based traversal achieves QPS = 17.11–15.84 for the two best configurations at mAP = 0.259–0.256, versus Grid Search's FaissHNSWFlat/IVFFlat configurations at mean QPS = 20.42 but mAP = 0.169. The Score contrast is stark: 4.44 and 4.05 for HNSW-based methods vs. 3.44 for Grid's sweep — confirming that high-QPS, low-recall configurations are penalized by the Score objective whenever mAP drops below a critical threshold. HNSW's navigable graph structure, with graph connectivity M=48 tuned to maintain sufficient Stage-1 recall, provides the best balance between ANN approximation quality and query throughput for 768-dimensional CLIP embeddings at k=200.

Query Difficulty-Weighted Optimization (QDS). The standard Score = mAP × QPS objective weights all queries equally, yet HOI queries vary substantially in retrieval difficulty: categories with high visual diversity (e.g., *ride horse*) span a wider region of CLIP space than visually compact categories (e.g., *hold cup*), making per-query AP more sensitive to index configuration. We introduce a *Query Difficulty Score* (QDS) that identifies structurally hard queries and upweights their contribution to the optimization objective. For query j, QDS combines two complementary signals:

$$H_j = -\sum_i p_{ji} \log p_{ji}, \quad \text{where } p_{ji} = \mathrm{softmax}(\mathrm{f}_{q_j}^\top \mathrm{f}_{d_i} / \tau)$$

measures *gallery entropy* — how uniformly the CLIP query mass spreads over the gallery (high H_j: CLIP is confused, many candidates look equally relevant); and

$$\mathrm{CPR}_j = \|\mathrm{f}_{q_j} - \boldsymbol{\mu}_j\|^2 / \sigma_j^2$$

measures the *centroid displacement ratio* — how far the text query embedding lies from the visual centroid $\boldsymbol{\mu}_j$ of its ground-truth positives (high CPR_j: the text query is not well-anchored to its class's visual region). Normalizing both to $[0, 1]$ and combining: $\mathrm{QDS}_j = \tilde{H}_j - \widetilde{\mathrm{CPR}}_j$, the per-query weight is $w_j = 1 + \tilde{I}_j \in [1, 2]$, and the weighted objective becomes $\text{QDS-mAP} = \sum_j w_j \cdot \mathrm{AP}_j \,/\, \sum_j w_j$.

When the LLM agent optimizes QDS-mAP × QPS, it discovers a configuration (FaissHNSWFlat, M=16, efSearch=500, k=150, α=0.75, n_r=10, *first*, constraint: object) that achieves Score = 4.66 on the *standard* unweighted metric — a +4.9% improvement over the LLM agent with standard objective (Score = 4.44) and the highest single-run Score across all methods on HICO-DET (Table 4). The QDS-guided optimizer reaches its best configuration at iteration 19, suggesting that difficulty-aware guidance accelerates convergence to a better optimum. Notably, the discovered configuration achieves higher QPS (19.17 vs. 17.11) by operating in a regime where the difficulty-weighted signal prefers configurations that maintain precision on hard queries rather than maximizing average recall.

6.7 Discussion

Theory-to-experiment alignment. Section 4 predicts that CLIP text queries should systematically outperform single reference image queries for HOI retrieval, because the text embedding lies near the visual centroid of the HOI class while any individual image is displaced by $\varepsilon_j > 0$. The system baselines in Table 3 provide a controlled empirical test: at identical k=2000 and FaissFlat exact search, CLIP text queries achieve mAP = 0.240 while DINOv2 image-based retrieval achieves mAP = 0.217 (−2.3 pp). This 9.6% relative deficit in favor of text is consistent with the geometric prediction that image queries incur a positive bias $\varepsilon_j^2 > 0$ that text queries avoid. The advantage is expected to be largest for visually diverse HOI categories (large ε_j), where any single reference image is a poor proxy for the class distribution — precisely the setting that dominates HICO-DET's long-tailed structure.

LLM architectural reasoning. Beyond sample efficiency, the LLM agent demonstrates a qualitatively different form of reasoning from Bayesian and non-adaptive baselines: it explicitly reasons about the interaction structure of the pipeline before proposing configurations. During the exploitation phase, the agent selects efSearch=64 < k=200 — a configuration that deliberately accepts ANN recall loss — while simultaneously compensating by raising α to 0.80. This cross-stage compensatory reasoning, in which a relaxed upstream constraint is offset by a tighter downstream parameter, requires modeling the pipeline as an integrated system rather than a product of independent components. Optuna TPE's univariate marginal density model and Random Search's independence assumption are both structurally incapable of encoding this dependency, explaining why both methods identify high-QPS configurations with weaker reranking (QPS = 20.47, mAP = 0.200 for Random's best) rather than the balanced operating point the LLM discovers.

Limitations. The present evaluation spans three datasets (HICO-DET at 47K, GLDv2 at 762K, SIFT1M at 1M vectors) covering both semantic and non-semantic retrieval, but all corpora are considerably smaller than production-grade vector databases at billion scale. Generalization to billion-scale corpora, where memory locality, IVFFlat's coarser quantization effects, and network storage latency introduce coupling dimensions not encountered in the present experiments, remains an open question. Statistical confidence across seeds has been quantified for LLM, Optuna TPE, and Random Search (3 seeds each); extending this to Grid Search and additional LLM families remains future work. Finally, the LLM agent's behavior is specific to MiniMax-M2.1 via OpenRouter; different model families may differ in convergence speed, architectural priors, and sensitivity to the five feedback class labels used in the per-iteration diagnostic.

7 CONCLUSION

We addressed two intertwined problems that arise when deploying vision-language embeddings in a production vector database: the absence of a semantically demanding retrieval benchmark that tests compositional understanding, and the lack of principled methods for jointly optimizing ANN index accuracy and throughput.

On the benchmark side, we introduced the first text-query image retrieval benchmark derived from HICO-DET — 47,776 images, 600 HOI categories, and 90,641 ground-truth positive pairs — and demonstrated that HICO-DET's compositional structure creates a retrieval task that is qualitatively harder than scene- or object-level benchmarks: even exhaustive exact-nearest-neighbor search with modality fusion reaches only mAP = 0.276, exposing a persistent semantic ceiling that reflects genuine language–vision alignment difficulty rather than any indexing limitation.

On the retrieval pipeline side, we proposed a two-stage architecture that pairs CLIP ViT-L/14 ANN retrieval with DINOv2 ViT-L/14-reg4 visual reranking at deployment cost of a single index. DINOv2 reranking improves mAP by 3.65 percentage points (+15.2% relative) over CLIP-only retrieval at under 1% additional query latency. We provided a formal geometric account of why text queries outperform reference image queries on compositionally structured datasets: a CLIP text embedding occupies the centroid of the HOI category's visual cluster, while any single reference image is displaced from that centroid by $\varepsilon_j > 0$, contributing a bias term ε_j^2 that increases expected L2 distance to every class member. This bias is largest for visually diverse categories with large within-class spread — precisely the long-tailed regime that dominates HICO-DET.

On the optimization side, we identified that ANN retrieval search spaces have hard conditional dependencies — engine type gates valid parameters, and Stage-1 search depth couples to Stage-2 fusion weight — that cause parameter-independent methods (Bayesian TPE with univariate marginals, random sampling) to fail structurally on this class of problems. We presented a phase-aware LLM agent that conditions each proposal on the full history of (configuration, Score) pairs and reasons about engine architecture, graph connectivity, search depth, and fusion weight as an integrated system, discovering cross-stage compensatory configurations that neither Bayesian nor non-adaptive baselines locate within a 30-iteration budget. The agent reaches Score $\approx$ 3.97 within 6 iterations — a threshold that requires 27 iterations for Optuna TPE and 28 for Random Search, saving approximately 3 hours and 3.5 hours of compute respectively. Across three independent seeds, the LLM agent achieves a mean Score = 4.44 (mAP × QPS), a 2.5× improvement over the CLIP FaissFlat baseline and 13.9× over brute-force CLIP+DINOv2 reranking, at only 2.1 percentage points of mAP below the exact-search upper bound — a 14.7× throughput gain (QPS = 16.98 vs. 1.15) that brute-force cannot approach.

Across three datasets, our evaluation tells a consistent and theoretically grounded story. On HICO-DET, the LLM agent achieves the highest mean Score (4.44), converging to ~95% of its best in ~13 iterations on average versus ~20 for Optuna TPE — and leads by +9.7% over Optuna TPE, +10.4% over Random Search, and +3.3% over GP-BO (VDTuner-style). On GLDv2 (762K landmark images, 16× scale), the LLM agent again ranks first (mean Score = 3.66) and reaches near-optimal quality in ~1–2 iterations, demonstrating that LLM architectural priors transfer across scale and retrieval modality. On SIFT1M — a structurally simpler, single-stage HNSW benchmark with no cross-stage coupling — all four methods (GP-BO excluded) converge within 2.1% of each other, confirming that the agent's advantage scales with the coupling complexity of the configuration landscape and is not an artifact of a favorable benchmark.

Looking forward, three directions are most immediate. First, extending the multi-seed evaluation (currently 3 seeds for all adaptive methods — LLM, Optuna TPE, GP-BO, and Random Search) to Grid Search and to additional LLM model families will quantify whether the sample-efficiency advantage is model-specific or a general property of history-conditioned optimization in coupled search spaces. Second, scaling to billion-scale corpora — where memory locality, quantization effects, and IVFFlat sharding introduce additional coupling dimensions not present at 47K or 762K scale — will test the limits of LLM architectural reasoning and motivate extensions to the phase structure and stagnation detection components. Third, the LLM agent framework is not specific to HNSW or vector databases: any system where a small number of structured parameters governs a measurable accuracy–throughput objective — including learned index structures, database query planners, and neural architecture selection — is a candidate application. We release the benchmark, pipeline, and optimizer code to support reproducibility and to encourage the community to evaluate compositional retrieval beyond object- and scene-level tasks.

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